Towards Cocoon: A Simulation Scaffold for Citizen-Co-Created Low-Height Distributed MIMO Access Networks

Gilles Callebaut *,
* KU Leuven, Belgium

***Abstract*—This manuscript is structured as a concept-plus-simulation paper for the Cocoon vision. The current draft focuses on the simulation scaffold built in `COCOON-Sionna`: an outdoor distributed MIMO (D-MIMO) pipeline that generates low-height candidate access points (APs), simulates user mobility, computes path-based channel state information (CSI), and compares movable-AP placement strategies over a common urban scenario. The interpretation of the generated metrics is intentionally deferred. Instead, this draft provides the paper structure, figure hooks, and the core technical framing needed to start incorporating results from committed simulation outputs.** *gilles* **Replace this abstract with quantitative claims once the first stable result set is frozen.**

***Index Terms*—cell-free systems, distributed MIMO, low-height access points, access-point placement, urban mobility, Sionna RT**

I. Introduction

The Cocoon project targets a wireless access paradigm in which the network adapts to the fluidity of the city rather than assuming a static infrastructure and a stationary demand profile. The motivating hypothesis from the project proposal is that a practical cell-free (CF) network becomes more feasible when low-height infrastructure can be co-created, repositioned, and activated only when the measured channel conditions justify it. In that view, the citizens are not only traffic sources but also anchors for sensing, mounting, and reconfiguring the surrounding network.

This manuscript narrows that broader vision to a first simulation paper. The current code base already supports a concrete subset of the Cocoon idea: low-height wall-mounted candidate access points (APs), pedestrian mobility, path-based channel state information (CSI) extraction, and repeated placement decisions over relocation windows in an outdoor urban scene. That subset is sufficient to establish a paper scaffold around environment geometry, strategy comparison, and performance figures, while deferring citizen-study interpretation and experimental validation to later versions.

The draft is therefore structured around three practical contributions that can be filled incrementally as results mature:

- a concise technical framing of Cocoon as a low-height, movable-AP, citizen-co-created access-network concept,

Working manuscript scaffold for the COCOON project. Update title, funding text, and author list before submission.

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Missing figure asset

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Rebuild or copy the expected output before finalizing the manuscript.

Fig. 1: Working concept figure for Cocoon. Replace `figures/cocoon_concept.pdf` with a redrawn version of the proposal illustration before submission.

- a manuscript-ready description of the Rabot outdoor simulation pipeline implemented in `COCOON-Sionna`, and
- stable figure hooks for geometry, signal-to-interference-plus-noise ratio (SINR), and relocation-cost results generated from committed `outputs/...` assets.

gilles Later pass: replace the generic contribution bullets with submission-grade claims and add citations to the proposal-aligned background literature.

II. Scope Within Cocoon

The full Cocoon proposal spans citizen interaction with the wireless network, channel assessment, architecture design, and proof-of-concept validation. The present paper should be positioned more narrowly as an early validation layer that sits mainly between the channel-adaptability and architecture-and-algorithms work packages. In proposal terminology, it is closest to the simulation-facing parts of WP2 and WP3: dynamic channel-aware adaptation, low-height AP/contact service point (CSP) placement, and the algorithmic consequences of allowing physical repositioning.

The main modeling simplification is that co-creation is represented here through an abstract movable infrastructure layer rather than through a user study or a deployment trial. Candidate mounting locations stand in for positions that could

later be offered to citizens, and the placement scheduler stands in for the future tool chain that would notify participants where the next best CSP constellation should be installed.

This scope should be stated explicitly in the manuscript because it clarifies both the ambition and the limits of the current evidence:

- the paper validates a simulation methodology and a comparison workflow, not the social feasibility of citizen participation,
- the paper studies a Rabot outdoor scenario, not yet a multi-site or multi-season campaign,
- the paper compares placement strategies over a common CSI-derived objective, not yet a full end-to-end network stack.

gilles Later pass: add one paragraph that maps each paper section to the relevant Cocoon work-package objective once the target venue is fixed.

III. System Model

We consider a downlink distributed MIMO (D-MIMO) system with active APs $\mathcal{A} = \{1, \ldots, M\}$ and scheduled user equipments (UEs) $\mathcal{U} = \{1, \ldots, K\}$. Let N_m denote the number of transmit ports at AP m and $N_{r,k}$ the number of receive ports at UE k. The per-AP channel block is

$$\mathbf{H}_{k,m} \in \mathbb{C}^{N_{r,k} \times N_m}, \tag{1}$$

and the full distributed channel observed by UE k is

$$\mathbf{H}_k = [\mathbf{H}_{k,1}\ \mathbf{H}_{k,2}\ \cdots\ \mathbf{H}_{k,M}]. \tag{2}$$

Given a receive combiner $\mathbf{u}_k$, the effective channel becomes

$$\mathbf{h}_k^{\mathsf{H}} = \mathbf{u}_k^{\mathsf{H}} \mathbf{H}_k. \tag{3}$$

Stacking all users yields the effective distributed channel matrix $\mathbf{H}_{\text{eff}}$. The current pipeline applies a least-squares pseudo-inverse zero-forcing (ZF) precoder on $\mathbf{H}_{\text{eff}}$, followed by per-stream power allocation under a total AP power budget.

For UE k, the instantaneous downlink SINR is evaluated as

$$\text{SINR}_k^{\text{DL}} = \frac{p_k \left|\mathbf{h}_k^{\mathsf{H}} \mathbf{f}_k\right|^2}{\sum_{i \in \mathcal{U},\, i \neq k} p_i \left|\mathbf{h}_k^{\mathsf{H}} \mathbf{f}_i\right|^2 + \sigma^2}, \tag{4}$$

where $\mathbf{f}_i$ is the precoding vector for stream i, p_i its allocated power, and σ^2 the thermal-noise term. The implementation keeps the interference term explicitly and exports desired-signal, interference, and noise power terms for each evaluated user snapshot.

The ergodic sum rate (ESR) used later in the results section is derived from the same per-snapshot SINR values. For snapshot n with scheduled-user set $\mathcal{U}_n$, the postprocessing layer computes

$$\text{ESR}_n = \sum_{k \in \mathcal{U}_n} \log_2\left(1 + \text{SINR}_{k,n}^{\text{DL}}\right). \tag{5}$$

This manuscript uses the SINR- and ESR-based figures as reporting views only. The text below does not interpret any numeric outcome.

A. Placement Strategies

The placement layer compares three strategies under a common AP budget:

- `central_massive_mimo`: one rooftop proxy AP is placed on the building whose rooftop representative point is nearest the area center; the proxy is oriented toward the area center. This is a normalized central reference, not a full single co-located-array placement model.
- `distributed_fixed`: a distributed set of wall-mounted candidate positions is sampled once and kept fixed over the full trajectory.
- `distributed_movable`: the same number of distributed APs is reselected at each relocation window from the wall-mounted candidate pool.

Let $\mathcal{C}$ denote the wall-mounted candidate pool and let $\mathcal{S}_w \subseteq \mathcal{C}$ denote the selected movable-AP subset in relocation window w. The current Rabot manuscript focuses on the case $|\mathcal{S}_w| = N$ with no always-on distributed APs, so the distributed comparison isolates the effect of changing the movable subset over time.

B. Relocation Heuristic

For each relocation window, the movable strategy scores candidate subsets from local CSI. Historical measurements are retained through exponential decay. If a local CSI sample was observed at time t_k and the relocation decision is taken at time t_{now}, then the weighting term is

$$w_k = \exp[-\lambda\,(t_{\text{now}} - t_k)], \tag{6}$$

with λ the configured decay rate.

The current relocation heuristic can be described in four steps:

1) Generate low-height wall-mounted candidate positions from building geometry.
2) Collect per-snapshot AP-to-UE CSI over the simulated trajectories.
3) Aggregate local candidate scores with exponentially decayed historical weighting so recent CSI contributes more strongly than stale measurements.
4) Select the subset whose local weighted percentile objective is best within the configured heuristic search budget.

The manuscript should describe these mechanics, not the realized numeric ranking. Any claim about which strategy wins belongs only after the figures are regenerated and frozen.

IV. Simulation Setup

The current manuscript scaffold is instantiated with the Rabot outdoor scenario defined in `scenarios/rabot.yaml`. The scene is derived from an OpenStreetMap (OSM) boundary around the KU Leuven Gent Campus Rabot environment. Candidate distributed-AP positions are generated along building walls at low height, one rooftop proxy is reserved for the central reference strategy, and user mobility is simulated over the outdoor walkable area with mixed pedestrian and bicycle profiles.

TABLE I: Key Rabot scenario parameters used by the current manuscript scaffold.

Parameter	Value
Carrier frequency	3.5 GHz
Bandwidth	100 MHz
User height	1.5 m
Number of users	10
Simulation horizon	600 s
Time step	10 s
Distributed APs per strategy	6
Fixed distributed APs	0
Candidate wall height	1.5 m
Candidate wall spacing	18 m
Minimum candidate spacing	12 m
Candidate wall offset	0.25 m
Relocation window	60 s
Historical CSI decay rate	0.011 55 s^{-1}
Compared strategies	central proxy, fixed distributed, movable distributed
Coverage maps	disabled in current Rabot config

TABLE II: Placement-strategy description used throughout the article. Fill the final implementation notes after rerunning the outputs, but keep the definitions themselves stable.

Strategy	Description
`central_massive_mimo`	Single rooftop proxy on the building nearest the area center, oriented toward the center. Use as central reference only.
`distributed_fixed`	One distributed subset is selected from wall candidates and kept unchanged for the full run.
`distributed_movable`	The distributed subset is reselected every relocation window using local CSI with exponential decay over historical samples.

The simulation workflow used by the paper matches the repository workflow:

1) build the outdoor scene and walk graph from OSM,
2) generate wall-mounted candidate APs and the single rooftop proxy site,
3) simulate user trajectories over the walkable outdoor area,
4) compute path-based CSI and explicit SINR terms,
5) compare the central, fixed-distributed, and movable-distributed strategies,
6) export reusable result files,
7) generate manuscript figures in a separate postprocessing step.

This section should stay descriptive and reproducible. It should document how the figures are produced, not what the figures mean.

The manuscript should also state the output contract because the figures in Sections V to VII are linked directly to generated artifacts. In the present workflow, scene-layout figures live in `outputs/rabot/`, while SINR, ESR, and schedule figures live in `outputs/rabot/postprocessing/`. The article should treat these files as placeholders until the simulation is rerun.

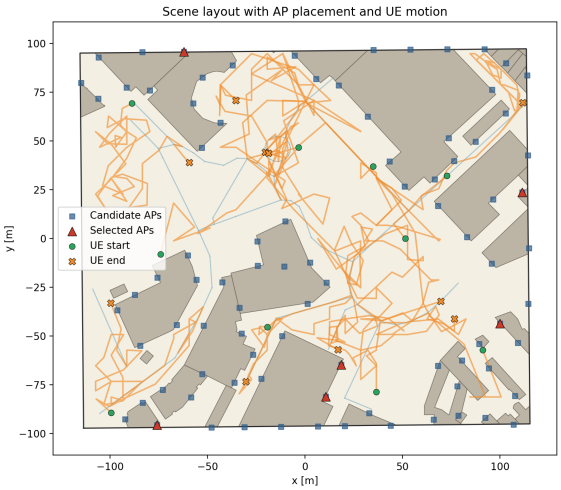
gilles Optional later addition: add one compact pipeline schematic once the figure inventory is frozen.

V. Scenario Geometry and Mobility Results

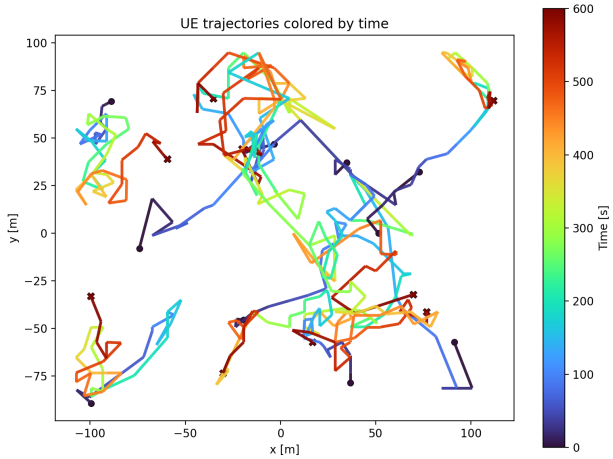
The first result block should orient the reader in the simulated environment before any performance metric is discussed. The goal is to show where candidate mounting locations exist, how the selected constellation relates to the urban geometry, and how the user trajectories populate the scene over time.

The text accompanying Fig. 2 should stay factual in this draft. It can describe the modeled environment, the spread of the trajectories, and the low-height infrastructure assumption without yet claiming why one region or path is more favorable than another.

gilles Later pass: add 2–3 sentences on how the candidate-wall generation and user-motion spread affect the interpretation of later SINR and schedule figures.



(a) Top-down Rabot scene layout with candidate sites, selected sites, and simulated mobility overlays.



(b) Trajectory colormap used to visualize the temporal spread of user motion through the walkable space.

Fig. 2: Geometry-facing result hooks for the Rabot scenario. These figures should remain the first visual anchor for the simulation setup before performance interpretation begins.

VI. SINR- and ESR-Oriented Results

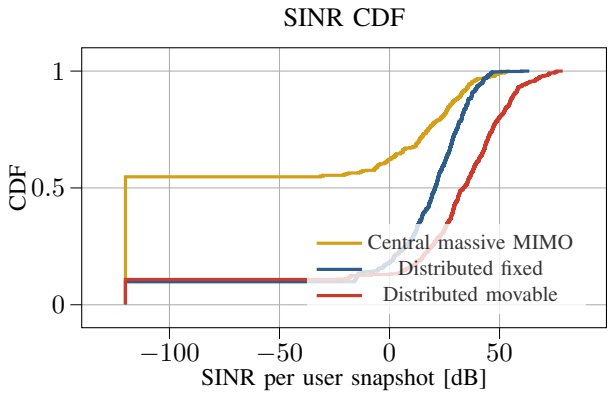
This section should collect the radio-performance figures produced by postprocessing, but it should not interpret them until the Rabot scenario is rerun. The role of the present draft is to anchor each figure to a precise artifact path and to explain what each placeholder represents.

The final prose in this section should answer only three structural questions: what metric is plotted, which exported file produced the figure, and which placement strategies appear in the comparison. Any statement about gains, fairness, outage reduction, or relocation benefit should be added only after the figures are regenerated and verified.

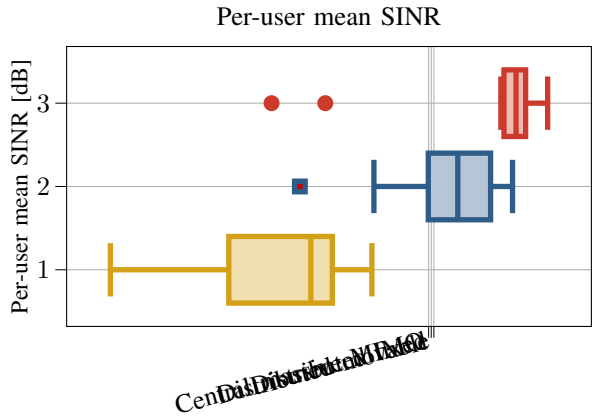
gilles Later pass: decide whether `users_below_threshold` and `esr_time_conditioned_cdf` should remain in the main paper or move to an appendix.

TABLE III: Placeholder map between manuscript figures and post-processed artifacts. Replace only the prose around the table after rerunning the pipeline.

Manuscript placeholder	Expected postprocessed artifact
Global SINR distribution	`postprocessing/sinr/sinr_cdf_publication.te`
Per-user mean boxplot	`postprocessing/sinr/per_user_mean_sinr_boxp`
ESR time series	`postprocessing/sinr/esr_timeseries.tex`
ESR CDF	`postprocessing/sinr/esr_cdf.tex`
Optional worst-user view	`postprocessing/sinr/worst_user_sinr_timese`

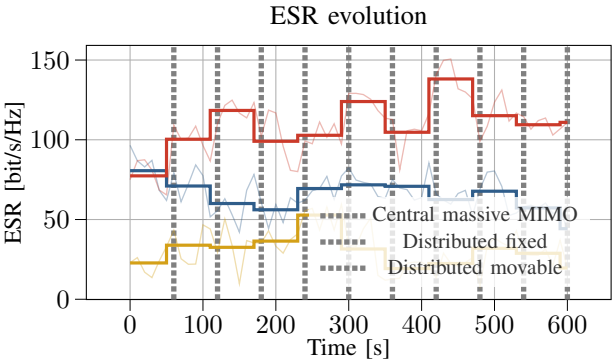


(a) Empirical SINR distribution across all user snapshots and compared placement strategies. Use the postprocessed `sinr_cdf_publication` asset here.

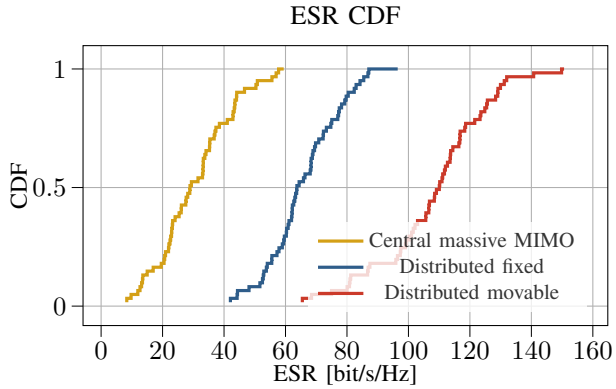


(b) Per-user mean SINR boxplot across strategies. Use this figure to summarize the distribution of user-level averages without adding interpretation yet.

Fig. 3: First pair of radio-performance placeholders. The left panel is the global SINR distribution, and the right panel is the per-user mean boxplot.



(a) ESR over time. Keep the relocation markers and step-like window averages visible, but defer any claim about increase or decrease until the outputs are rerun.



(b) ESR cumulative distribution function (CDF) across all evaluated snapshots. Use the `esr_cdf` artifact directly.

Fig. 4: Second pair of radio-performance placeholders. These panels report the ESR time evolution and the global ESR distribution.

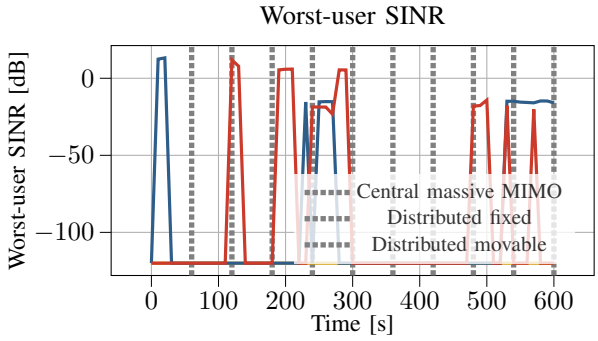


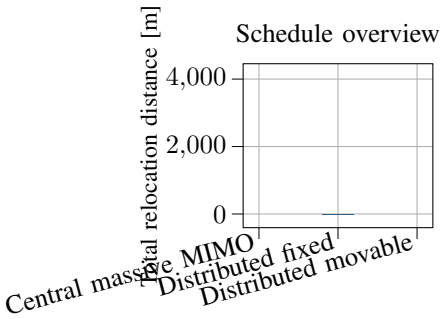
Fig. 5: Optional supporting placeholder for worst-user SINR over time. Keep this panel if temporal robustness of the weakest user becomes part of the final narrative.

VII. Relocation and Practicality Results

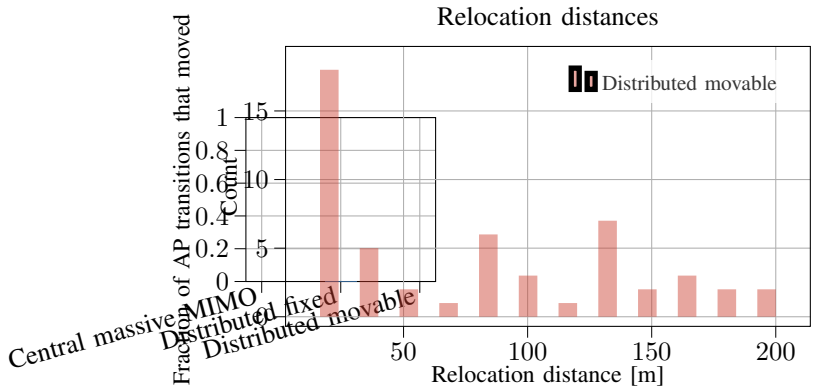
Because Cocoon is motivated by repositionable low-height infrastructure, the manuscript should not stop at radio-quality figures. The scheduling outputs quantify how often the movable APs change position and how much displacement each strategy induces. This is the current paper's first bridge between radio performance and physical practicality.

The final prose here should connect Fig. 6 to the Cocoon motivation without overselling what the current simulations can prove. At this stage, the section only needs to establish that the repository already provides reproducible schedule summaries and that these figures belong in the manuscript alongside the SINR results.

gilles Later pass: discuss whether relocation frequency and total distance should be reframed as user burden, maintenance overhead, or implementation cost depending on the target venue.



(a) Strategy-level overview of total relocation distance and move fraction.



(b) Distribution of relocation distances across observed AP moves.

Fig. 6: Relocation-cost figures used to interpret how much physical infrastructure movement is required by the compared placement strategies.

VIII. Discussion and Current Limits

This draft deliberately stops short of a full interpretation layer. Still, the manuscript should state the boundaries of the current evidence clearly so that later quantitative claims remain anchored to what the simulation pipeline actually models.

The main limitations that should remain visible in the first paper version are the following:

- Only a single outdoor scenario is wired into the current manuscript scaffold.
- The `central_massive_mimo` comparison point is a rooftop proxy, not a full placement model for one true co-located massive-multiple-input multiple-output (MIMO) array.
- The current Rabot configuration disables coverage-map exports, so the paper centers on trajectory-level SINR and schedule metrics.
- The present draft is figure-structured but interpretation-light by design; the reported placeholders must be refreshed after rerunning the pipeline.
- Citizen participation is abstracted as movable candidate positions and scheduler actions rather than being validated experimentally.
- The present text does not yet discuss sensitivity to seeds, scenario perturbations, or broader deployment costs.

The next analysis pass should therefore answer at least three questions: how stable the reported gains are across seeds and scenarios, how strongly the conclusions depend on the chosen SINR objective, and how the relocation metrics should be interpreted in a real co-created deployment process.

gilles Later pass: split this section into limitations and future work if the venue expects a more formal discussion structure.

IX. Conclusion

This manuscript scaffold turns the current `COCOON-Sionna` repository into a paper-ready IEEE structure for a Cocoon concept-plus-simulation study. The draft already separates motivation, scope, system model, simulation setup, geometry results, SINR results, and relocation results, while keeping all figure references tied to centralized path macros.

The immediate next step is to replace the placeholder interpretation text with the first stable Rabot result set, freeze the concept figure, and populate the strategy summary table from the committed postprocessing artifacts. After that, the paper can expand toward richer scenario comparisons, coverage maps, and eventually the measurement and proof-of-concept layers envisaged in the broader Cocoon program.

Acknowledgment

gilles Add funding statement, collaborators, and infrastructure acknowledgments here.